

SHAMAL MUSTHAF A

Data Scientist | GenAI / LLM Engineer | AI Engineer | Data Engineering | BI

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PROFESSIONAL SUMMARY

Data Scientist and AI Solutions Builder with 2+ years of experience building **Generative AI** applications, **RAG** knowledge systems, **LLM**-powered chatbots, backend AI services, cloud **ETL** pipelines, **BI** dashboards, **machine learning** models and **workflow automation**. Hands-on with **Python**, **FastAPI**, **LangChain**, **FAISS**, **vector search**, **Azure OpenAI**, **OpenAI API**, **Gemini API**, **AWS Glue**, **Amazon S3**, **Redshift Serverless**, **Amazon RDS**, **SQL**, **Power BI**, **Tableau** and **Alteryx**. Strong in converting business requirements into production-oriented AI tools, APIs, analytics pipelines, dashboards and **automation** workflows with measurable impact.

CORE TECHNICAL SKILLS

Programming & Data Science: Python, SQL, Pandas, NumPy, scikit-learn, XGBoost, TensorFlow basics, data wrangling, exploratory data analysis, **feature engineering**, **model evaluation**, **classification**, **regression**, **anomaly detection**

Generative AI / LLM Engineering: RAG, LangChain, FAISS, vector databases, embeddings, chunking, **semantic search**, **prompt engineering**, **prompt templates**, AI agents, **tool calling**, MCP, AI summarization, chatbot workflows, conversational AI

AI Platforms & NLP: Azure OpenAI, OpenAI API, Gemini API, Vertex AI, Dialogflow CX, OpenAI Whisper, NLP, transcript analytics, text summarization, information retrieval, context-aware Q&A

Backend / Full Stack: FastAPI, REST APIs, NestJS, React, Next.js, TypeScript, JavaScript, Streamlit, **API integration**, authentication, OTP workflows, file upload workflows, **SQLModel**

Cloud / Data Engineering: AWS Glue, Amazon S3, Redshift Serverless, Amazon RDS, MySQL, PostgreSQL, **SQL Server**, **ETL/ELT**, **batch processing**, **incremental loads**, **watermark logic**, **data migration**, validation, **reconciliation**

BI / Analytics / Automation: Power BI, Tableau, DAX, Alteryx Designer, **dashboarding**, **KPI reporting**, **data modeling**, **business intelligence**, **workflow automation**, Git, Docker basics, Agile delivery, technical documentation

Computer Vision: **face recognition** evaluation, side-face image generation, **dataset creation**, image generation research, vulnerability analysis of recognition algorithms

PROFESSIONAL EXPERIENCE

Data Scientist / AI Solutions Builder | Truwave Software LLC, Madurai | Aug 2023 - Present

- Built enterprise **GenAI** and **LLM**-based solutions using **Azure OpenAI**, **OpenAI/Gemini APIs**, **LangChain**, **FAISS** and **Python** for **retrieval-augmented generation**, knowledge search, chatbot workflows and AI-assisted information access.
- Designed and implemented **RAG** knowledge base systems with document ingestion, chunking, **embeddings**, vector retrieval, **prompt templates** and conversational response generation for context-aware Q&A.
- Developed backend AI services in **FastAPI** and **NestJS** for chatbots, intelligent search, file processing, ticket summarization, OTP flows and API-based **automation** workflows integrated with **React/Next.js** applications.
- Created AI ticket summarization and support-review workflows using **Gemini API** and structured ticket data, reducing manual review effort by about 40% and improving triage speed.
- Built **AWS** analytics pipeline from **MySQL RDS** to **AWS Glue**, **Amazon S3**, **Redshift Serverless** and **Power BI** using transformation layers and **watermark**-based incremental loading, reducing downstream reporting data volume by about 99%.
- Designed **SQL/Python data migration** workflows using raw, staging, validation and target layers with legacy ID preservation, ID mapping, parent-child dependency checks, **reconciliation** and rollback planning.
- Delivered **Power BI** and **Tableau** dashboards for billing, meter usage, call-center analytics and operational **KPI reporting** with clean data models, **DAX** measures, filters, drilldowns and stakeholder-ready views.
- Automated data preparation and reporting workflows with **Python**, **SQL** and **Alteryx Designer**, reducing recurring processing effort by about 30% and improving repeatability of analytics delivery.
- Worked on water meter usage analytics across hourly, daily, monthly and yearly granularity; created features, quality checks, **anomaly detection** and leak/no-usage detection logic to support low-false-positive alerting.
- Integrated **Dialogflow CX** with **Gemini/OpenAI** services for conversational assistant flows, OTP verification logic, natural-language interfaces and business **workflow automation**.

Data Analyst Intern | Truwave Software LLC, Madurai | Jan 2023 - Apr 2023

- Cleaned, transformed and analyzed call-center and operational datasets using **Python**, **SQL**, **Alteryx** and BI workflows for reporting and performance analysis.
- Built **Power BI** dashboards to track call performance, customer satisfaction signals, service trends and operational KPIs.
- Explored transcript analytics using **OpenAI Whisper** and **NLP** techniques to convert unstructured call data into measurable insights and reporting-ready features.

SELECTED PROFESSIONAL PROJECT HIGHLIGHTS

- **Enterprise RAG Knowledge Base:** Built a retrieval-grounded AI assistant over business knowledge sources with document ingestion, **embeddings**, **vector search**, prompt workflows and conversational Q&A for faster information access. Tech: **Python**, **LangChain**, **FAISS**, **Azure OpenAI/OpenAI**, **FastAPI**.
- **RAG Chatbot Assistant:** Developed chatbot flows with context retrieval, conversational memory, **OpenAI/Gemini** integration and API-backed responses for business users. Tech: **Python**, **Gemini/OpenAI**, **LangChain**, **FastAPI**, **Streamlit**.
- **AI Ticket Summarization System:** Automated support-ticket summarization and search workflows to convert long ticket history into short actionable summaries and reduce manual review workload. Tech: **Gemini API**, **NestJS**, **REST APIs**, **SQL**.
- **AWS ETL & BI Pipeline:** Created cloud data pipeline from **MySQL RDS** to **AWS Glue/S3** to **Redshift Serverless** and **Power BI** with transformation logic, **incremental loads** and reporting-ready output tables. Tech: **AWS Glue**, **S3**, **Redshift**, **RDS**, **SQL**, **Power BI**.
- **Water Leak / Usage Anomaly Detection:** Built hourly, daily, monthly and yearly water-meter analytics with rolling features, no-usage checks, anomaly scoring and data-quality rules to support leak detection. Tech: **Python**, **Alteryx**, **Power BI**, **SQL**, **anomaly detection**.
- **Power BI Billing Dashboard:** Modeled billing and usage datasets, built **DAX** measures, slicers, filters, drilldowns and dashboard views for business monitoring and analysis. Tech: **Power BI**, **DAX**, **SQL**, **data modeling**.

- **Legacy Data Migration Framework:** Designed raw-to-staging-to-validation-to-target migration process with preserved legacy IDs, mapping tables, referential-integrity validation, **reconciliation** and sign-off checks. Tech: **Python, SQL, ETL, data validation**.
- **Call Center Analytics Dashboard:** Analyzed call-center data and transcript signals to create performance dashboards and operational insights from structured and unstructured data. Tech: **Power BI, Alteryx, NLP, Whisper**.

INDEPENDENT PROJECTS & AI AUTOMATION

- **Customer Churn Prediction App:** Developed an end-to-end churn prediction solution with **feature engineering, XGBoost model evaluation**, API serving and lightweight **React** frontend; achieved 84.6% ROC AUC. Tech: **Python, XGBoost, FastAPI, React**, Vercel.
- **Pipe Failure Prediction / Predictive Maintenance:** Built **predictive analytics** prototypes using asset, usage and operational features to support risk ranking, **anomaly detection** and maintenance prioritization. Tech: **Python, scikit-learn, SQL, Power BI**.
- **MCP Demo Server:** Built a **FastMCP** demo server with simulated Jira, GitHub and CloudWatch-style data to demonstrate tool-based AI querying and agentic workflow patterns. Tech: **FastMCP, Python, tool calling**, mock operational data.
- **AI Financial / Expense Review Automation:** Designed prompt-driven transaction review workflows that classify merchants, exclude transfers/investments, summarize spending and produce daily expense updates. Tech: Gmail data, **Python**-style logic, prompt workflows, categorization rules.
- **AI Job Application & Resume Automation:** Designed a safe-mode workflow for job discovery, resume tailoring, application-email drafting and Gmail draft preparation with manual approval before submission. Tech: **AI agents**, Gmail drafts, resume tailoring, safety checks.
- **Developer Productivity / Codex Workflow Support:** Designed AI-assisted workflows for code understanding, implementation planning, prompt reuse, documentation support and repeatable developer productivity tasks. Tech: Codex, structured prompts, AI-assisted workflows, documentation.
- **Next.js / FastAPI Auth System:** Built login and registration workflows with OTP-style authentication, **backend APIs**, database models and frontend pages for app onboarding. Tech: **Next.js, FastAPI, SQLAlchemy, PostgreSQL**.
- **Computer Vision Research:** Presented CIDA-23 research on side-face image generation for vulnerability analysis of face-recognition algorithms and **dataset creation** for robustness evaluation. Tech: **Computer Vision, face recognition**, dataset generation.

EDUCATION

M.Sc. Data Science | CHRIST University

Bachelor of Computer Applications (BCA) | Kannur University

CERTIFICATIONS & RESEARCH

- **Alteryx Designer** Core Certified
- **Presented research paper:** 'Side Face Image Generation for Vulnerability Analysis of **Face Recognition** Algorithms' at CIDA-23 national conference on Computational Intelligence and Data Analytics.
- **Microsoft AI learning:** Fundamental AI Concepts / Azure AI learning exposure

ATS KEYWORDS

Data Scientist, Generative AI Engineer, LLM Engineer, AI Engineer, Machine Learning Engineer, Python Developer, FastAPI, LangChain, RAG, Azure OpenAI, OpenAI API, Gemini API, Vertex AI, FAISS, Vector Database, Embeddings, Semantic Search, Prompt Engineering, AI Agents, Tool Calling, MCP, NLP, OpenAI Whisper, XGBoost, scikit-learn, Anomaly Detection, Predictive Modeling, AWS Glue, Amazon S3, Redshift Serverless, Amazon RDS, SQL, MySQL, PostgreSQL, SQL Server, ETL, ELT, Data Migration, Data Validation, Reconciliation, Power BI, Tableau, DAX, Alteryx, Automation, Dashboarding, Backend APIs, REST APIs, React, Next.js, NestJS.